

Milestone 3: Source Scanner Identification using CNNs

1. Objective

The objective of this milestone was to develop a Convolutional Neural Network (CNN) capable of identifying the source scanner of a document directly from raw image data. Unlike Milestone 2, which relied on handcrafted features like FFT and LBP, this deep learning approach allows the model to automatically learn unique "fingerprints" (micro-textures) left by different scanner sensors.

2. Data Preprocessing & Memory Optimization

Due to the high resolution of the scanned images (512 X 512), initial training attempts caused system RAM crashes. To solve this, two critical preprocessing steps were implemented:

- **Image Patching:** Instead of processing the full image, the model was trained on 64x64 center patches. This reduced memory consumption by approximately 64 times.
 - **Label Alignment:** The dataset labels were converted from 1-based indexing (1–11) to 0-based indexing (0–10) to ensure compatibility with the Softmax output layer.
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3. Model Architecture

A Sequential CNN was designed with the following layers to capture forensic signatures:

- **Convolutional Layers:** To extract spatial noise patterns and sensor artifacts.
 - **Max Pooling Layers:** To reduce dimensionality and focus on the most prominent texture features.
 - **Dropout (0.5):** To prevent overfitting and improve the model's ability to generalize to new data.
 - **Dense Layer (Softmax):** To classify the input into one of the 11 scanner categories.
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4. Experimental Results

The model was trained incrementally across 18 batches of data. After resolving initial label mismatches, the CNN achieved the following performance:

- **Final Accuracy:** 96%.
 - **Observations:** The model showed high precision for most scanner classes, demonstrating that deep learning can effectively capture scanner-specific noise without manual feature extraction.
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5.Conclusion & Comparison

Metric	Milestone 2 (Random Forest)	Milestone 3 (CNN)
Approach	Handcrafted Features (FFT/LBP)	Automated Feature Learning
Input	11 Statistical Features	Raw 64×64 Image Patches
Accuracy	~98.03%	96%
